CRITICAL CARE MEDICINE · MIDTERM · LECTURE 4 · QUESTION BANK

Pulmonary Edema

الارتشاح الرئوي — بنك الأسئلة

34	12	13
Past exam questions	NCLEX-style	Flashcards

المصادر اللي اتجمعت منها أسئلة المحاضرة دي:

كويزات سابقة (طب حالات حرجة) · أسئلة سابقة

جنب كل سؤال مصدره. لو السؤال اتكرر في كذا امتحان مكتوب كل مرة بمصدرها، وعليه علامة "تكرر". الإجابات كلها في آخر الملف.

اللَّهُمَّ لا سَهْلَ إِلَّا ما جَعَلْتَهُ سَهْلًا، وأنتَ تَجْعَلُ الحَزْنَ إذا شِئْتَ سَهْلًا — صلِّ على النبي ﷺ

Part A · Past Exam Questions

أسئلة الدفعات السابقة

1 كويزات سابقة (طب حالات حرجة)

Pulmonary edema is characterized by which of the following? 1. Frothy secretions 2. Induced by heart failure 3. Induced by bronchial asthma 4. Tachypnea

a. 1,3,4
b. 1,2,4
c. 2,3,4

2 كويزات سابقة (طب حالات حرجة)

Which one of the following is a management option for pulmonary edema?

a. Performing a thoracoscopy
b. Administering specific antibiotic
c. Putting the patient on CPAP mode
d. Administering anticoagulant drugs

3 أسئلة سابقة

What is the most common cause of cardiogenic pulmonary edema?

a. Pneumonia
b. Acute pancreatitis
c. Left ventricular failure
d. Sepsis

4 أسئلة سابقة

Which of the following is NOT a direct lung injury that can lead to non-cardiogenic pulmonary edema?

a. Pneumonia
b. Sepsis
c. Inhalational injury
d. High altitude pulmonary edema

5 أسئلة سابقة

Which of the following is characteristic of pulmonary edema?

a. Sudden chest pain and loss of consciousness
b. Frothy sputum and shortness of breath
c. Pain in the lower extremities
d. Decreased lung sounds on auscultation

6 Which diagnostic test is used to assess hypoxemia and hypercapnia in pulmonary edema? أسئلة سابقة

a. Chest X-ray
b. Arterial Blood Gas (ABG)
c. Echocardiogram
d. CT Chest

7 What would you expect to see on a Chest X-ray (CXR) in a patient with pulmonary edema? أسئلة سابقة

a. Hyperinflated lungs with flat diaphragm
b. Increase in pulmonary parenchymal opacification
c. Ground glass opacification
d. Normal lung fields

8 Which of the following is most commonly associated with non-cardiogenic pulmonary edema? أسئلة سابقة

a. Myocardial infarction
b. Pneumonia
c. Hypertension
d. Congestive heart failure

9 Which is a typical symptom of pulmonary edema? أسئلة سابقة

a. Increased sputum production
b. Severe chest pain
c. Cyanosis and sudden collapse
d. Shortness of breath (dyspnea (

10 Which of the following is a characteristic finding in the physical examination of pulmonary edema? أسئلة سابقة

a. Bilateral wheezes without crepitations
b. Congested neck veins
c. Decreased respiratory rate
d. Unilateral reduced breath sounds

11 What is the role of BNP (B-type natriuretic peptide) in diagnosing pulmonary edema? أسئلة سابقة

a. Detects myocardial infarction
b. Indicates the stretch of the left atrium in case of heart failure
c. Detects bacterial infection
d. Measures oxygen levels in the blood

12 In the context of pulmonary edema, what is the purpose of using a Pulse Oximeter? أسئلة سابقة

a. To evaluate heart rhythm
b. To assess lung compliance
c. To measure oxygen saturation in the blood
d. To check for fluid buildup in the lungs

13 How is cardiogenic pulmonary edema differentiated from non-cardiogenic pulmonary edema? أسئلة سابقة

a. By using a chest X-ray
b. By using an echocardiogram
c. By measuring arterial blood gases
d. By performing a bronchoscopy

14 What is the most common cause of non-cardiogenic pulmonary edema? أسئلة سابقة

a. Pneumonia
b. Sepsis
c. Hypertensive crisis
d. Myocardial infarction

15 What medication is used to treat pulmonary edema by reducing fluid overload? أسئلة سابقة

a. Diuretics
b. Antibiotics
c. Steroids
d. Bronchodilators

16 Which treatment is used for severe pulmonary edema cases that require ventilatory support? أسئلة سابقة

a. Antihypertensive medication
b. Mechanical ventilation
c. Antibiotic therapy
d. Bronchodilator therapy

17 What does the term "frothy secretions" in pulmonary edema refer to? أسئلة سابقة

a. Blood-stained sputum
b. Mucous secretion mixed with air
c. Thick yellow sputum
d. Normal clear sputum

18 In the case of unstable hemodynamics due to pulmonary edema, which treatment is indicated? أسئلة سابقة

a. Diuretics
b. Defibrillation
c. Mechanical ventilation
d. Oxygen therapy

19 Which of the following findings on CT chest is characteristic of pulmonary edema? أسئلة سابقة

a. Hyperinflated lungs
b. Ground glass opacification
c. Consolidation
d. Pneumothorax

20 What is the key difference between cardiogenic and non-cardiogenic pulmonary edema? أسئلة سابقة

a. Cardiogenic edema responds to diuretics, whereas non-cardiogenic usually does not
b. Cardiogenic edema is caused by pneumonia, while non-cardiogenic is due to heart failure
c. Non-cardiogenic edema is treated with corticosteroids, cardiogenic is not
d. Non-cardiogenic edema typically requires mechanical ventilation, cardiogenic does not

21 Which of the following conditions is NOT a cause of non-cardiogenic pulmonary edema? أسئلة سابقة

a. Sepsis
b. High altitude pulmonary edema
c. Hypertension
d. Pneumonia

22 Which treatment is generally NOT used in the management of pulmonary edema? أسئلة سابقة

a. Oxygen therapy
b. Diuretics
c. Fluid overload management
d. Steroids

23 What is a common sign found on a CXR for patients with pulmonary edema? أسئلة سابقة

a. Increased lung parenchymal opacification
b. Hyperinflated lungs
c. Ground glass opacification
d. Pneumothorax

24 In which case is mechanical ventilation most likely required in pulmonary edema? أسئلة سابقة

a. Mild shortness of breath
b. Severe dyspnea with hypoxemia
c. Cough with little sputum production
d. No hypoxemia

25 Which of the following is NOT a recommended preventive measure for pulmonary edema? أسئلة سابقة

a. Regular check-ups with a cardiologist
b. Maintaining a healthy weight
c. Avoiding smoking
d. Consuming a high-salt diet

26 Which of the following is a key diagnostic tool for differentiating between cardiogenic and non- cardiogenic pulmonary edema? أسئلة سابقة

a. CT Chest
b. Echocardiogram
c. Arterial Blood Gas
d. Chest X-ray

27 What is the role of diuretics in the treatment of pulmonary edema? أسئلة سابقة

a. To reduce inflammation in the lungs
b. To increase oxygen saturation in the blood
c. To reduce fluid overload and relieve pressure on the heart
d. To decrease bronchospasm

28 Which of the following is a typical finding in a patient with cardiogenic pulmonary edema? أسئلة سابقة

a. Normal lung compliance
b. Increased lung interstitial fluid
c. Ground glass opacification
d. Unilateral chest pain

29 Which of the following conditions can lead to non-cardiogenic pulmonary edema? أسئلة سابقة

a. Myocardial infarction
b. Aspiration of gastric contents
c. Severe arrhythmias
d. Hypertension

30 What is the most appropriate action for a patient with pulmonary edema caused by sepsis? أسئلة سابقة

a. Diuretics to relieve fluid buildup
b. Treat the underlying infection with antibiotics
c. Administer mechanical ventilation
d. Administer corticosteroids

31 Which of the following is true about pulmonary edema treatment? أسئلة سابقة

a. Fluids are used to treat pulmonary edema
b. Oxygen therapy is indicated for all patients
c. Diuretics are contraindicated in non-cardiogenic pulmonary edema
d. Bronchodilators are the first-line treatment

32 In a patient with pulmonary edema and severe hypoxemia, what is the target SpO2 range? أسئلة سابقة

a. 80-85%
b. 88-92%
c. 95-100%
d. 75-80%

33 What is a key difference between ARDS and cardiogenic pulmonary edema? أسئلة سابقة

a. ARDS is primarily caused by heart failure, while cardiogenic edema is caused by infections
b. ARDS is characterized by bilateral infiltrates on CXR, which is not seen in cardiogenic edema
c. ARDS patients do not require intubation, while cardiogenic edema requires mechanical ventilation
d. ARDS responds to high-volume fluid resuscitation, unlike cardiogenic pulmonary edema

34 Which of the following findings would support a diagnosis of ARDS rather than cardiogenic pulmonary edema? أسئلة سابقة

a. Presence of pleural effusion
b. Presence of bilateral infiltrates on CXR
c. Increased oxygenation despite high PEEP
d. History of recent myocardial infarction

Part B · NCLEX-Style Practice

أسئلة على طريقة NCLEX (إضافة)

N1 The main mechanism of cardiogenic pulmonary edema is: NCLEX

a. Increased pulmonary capillary hydrostatic pressure from left ventricular failure
b. Increased capillary permeability from sepsis
c. Low oncotic pressure only
d. Airway obstruction

N2 Classic signs of acute pulmonary edema include (select all): SELECT ALL

a. Severe dyspnea and orthopnea
b. Pink frothy sputum
c. Bilateral fine crackles
d. Anxiety and diaphoresis
e. Bradypnea

N3 The best position for a patient with acute pulmonary edema is: NCLEX

a. Supine with legs elevated
b. High Fowler's with legs dependent
c. Prone
d. Trendelenburg

N4 Treatment of acute cardiogenic pulmonary edema includes (select all): SELECT ALL

a. Oxygen / CPAP or BiPAP
b. IV furosemide
c. Nitrates if BP allows
d. Rapid IV fluid boluses
e. Morphine (cautiously) in selected cases

N5 After IV furosemide, the nurse should monitor closely for: NCLEX

a. Hypokalemia and hypotension
b. Hyperkalemia
c. Hyperglycemia only
d. Weight gain

N6 A BNP level that is markedly elevated in a dyspneic patient suggests: NCLEX

a. Cardiac (heart failure) cause
b. Pneumonia
c. Asthma
d. Pulmonary embolism only

N7 Chest X-ray findings of cardiogenic pulmonary edema include (select all): SELECT ALL

a. Cardiomegaly
b. Kerley B lines
c. Bat-wing perihilar opacities
d. Pleural effusions
e. Hyperinflated lungs

N8 **Non-invasive CPAP helps in pulmonary edema by (select all):** SELECT ALL

a. Recruiting alveoli and improving oxygenation
b. Reducing preload
c. Reducing LV afterload
d. Increasing venous return
e. Reducing work of breathing

N9 **Causes of non-cardiogenic pulmonary edema include (select all):** SELECT ALL

a. Sepsis
b. Aspiration
c. High altitude
d. Near-drowning
e. Mitral stenosis

N10 **Nitroglycerin is withheld in pulmonary edema if:** NCLEX

a. Systolic BP < 90 mmHg
b. Heart rate is 90
c. SpO_2 is 92%
d. Patient is anxious

N11 **Indications for intubation in acute pulmonary edema include (select all):** SELECT ALL

a. Failure of NIV
b. Decreased level of consciousness
c. Severe respiratory acidosis or exhaustion
d. Hemodynamic instability
e. Mild dyspnea on exertion

N12 **Teaching to prevent recurrence of cardiogenic pulmonary edema includes (select all):** SELECT ALL

a. Low-salt diet and fluid restriction as ordered
b. Daily weight; report gain > 1–2 kg in a day
c. Take medications regularly
d. Stop diuretics when feeling well
e. Report increasing dyspnea or orthopnea

Part C · Quizlet Flashcards

كروت مراجعة سريعة

غطّي العمود الأبيض بورقة واختبر نفسك: شوف المصطلح وقول التعريف.

Pulmonary edema	Fluid accumulation in the lung interstitium and alveoli → impaired gas exchange.	**Cardiogenic**	LV failure, MI, valve disease, hypertensive crisis → raised hydrostatic pressure.
Non-cardiogenic	Increased permeability: sepsis (most common), pneumonia, aspiration, inhalation injury, high altitude.	**Symptoms**	Severe dyspnea, orthopnea, PND, frothy (pink) sputum, cough, anxiety.
Signs	Tachypnea, tachycardia, bilateral crackles ± wheeze, congested neck veins, cyanosis.	**Investigations**	ABG, pulse oximetry, CXR (↑ parenchymal opacity), CT (ground glass), ECG, BNP, echo.

Echo	Key test to differentiate cardiogenic from non-cardiogenic.
Management	Sit up, O_2 (SpO_2 88–92% target in severe hypoxemia), CPAP/NIV, diuretics, nitrates, treat cause; MV if failing.
Sepsis-induced	Treat infection with antibiotics; diuretics less useful.
ARDS vs cardiogenic	ARDS: bilateral infiltrates, normal heart size, no response to diuretics, refractory hypoxemia.

BNP	Released with ventricular / atrial stretch — high in cardiogenic.
Diuretics	Furosemide reduces fluid overload; watch K^+ and BP.
Not used	Steroids are not routine treatment; avoid high-salt diet.

Answer Key

الإجابات

1. b — 1,2,4
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

2. c — Putting the patient on CPAP mode
إجابة مقترحة (المصدر مفيهوش إجابة رسمية).

3. c — Left ventricular failure

4. b — Sepsis

5. b — Frothy sputum and shortness of breath

6. b — Arterial Blood Gas (ABG)

7. b — Increase in pulmonary parenchymal opacification

8. b — Pneumonia

9. d — Shortness of breath (dyspnea (

10. b — Congested neck veins

11. b — Indicates the stretch of the left atrium in case of heart failure

12. c — To measure oxygen saturation in the blood

13. b — By using an echocardiogram

14. b — Sepsis

15. a — Diuretics

16. b — Mechanical ventilation

17. b — Mucous secretion mixed with air

18. b — Defibrillation
الإجابة دي من الملف زي ما هي. الـ DC shock (cardioversion) بيكون لو عدم الاستقرار سببه arrhythmia.

19. b — Ground glass opacification

20. a — Cardiogenic edema responds to diuretics, whereas non-cardiogenic usually does not

21. c — Hypertension

22. d — Steroids

23. a — Increased lung parenchymal opacification

24. b — Severe dyspnea with hypoxemia

25. d — Consuming a high-salt diet

26. b — Echocardiogram

27. c — To reduce fluid overload and relieve pressure on the heart

28. b — Increased lung interstitial fluid

29. b — Aspiration of gastric contents

30. b — Treat the underlying infection with antibiotics

31. b — Oxygen therapy is indicated for all patients

32. b — 88-92%

33. b — ARDS is characterized by bilateral infiltrates on CXR, which is not seen in cardiogenic edema

34. b — Presence of bilateral infiltrates on CXR

N1. a — Increased pulmonary capillary hydrostatic pressure from left ventricular failure
Non-cardiogenic (ARDS) = increased permeability.

N2. a, b, c, d — Severe dyspnea and orthopnea; Pink frothy sputum; Bilateral fine crackles; Anxiety and diaphoresis
Tachypnea, tachycardia and hypoxemia.

N3. b — High Fowler's with legs dependent
Reduces venous return and eases breathing.

N4. a, b, c, e — Oxygen / CPAP or BiPAP; IV furosemide; Nitrates if BP allows; Morphine (cautiously) in selected cases
Fluids worsen the overload.

N5. a — Hypokalemia and hypotension
Also strict intake/output and daily weight.

N6. a — Cardiac (heart failure) cause
BNP is released when ventricles are stretched.

N7. a, b, c, d — Cardiomegaly; Kerley B lines; Bat-wing perihilar opacities; Pleural effusions
Hyperinflation is typical of COPD/asthma.

N8. a, b, c, e — Recruiting alveoli and improving oxygenation; Reducing preload; Reducing LV afterload; Reducing work of breathing
Positive intrathoracic pressure reduces venous return.

N9. a, b, c, d — Sepsis; Aspiration; High altitude; Near-drowning
Mitral stenosis raises left atrial pressure → cardiogenic.

N10. a — Systolic BP < 90 mmHg
It reduces preload and afterload but causes hypotension.

N11. a, b, c, d — Failure of NIV; Decreased level of consciousness; Severe respiratory acidosis or exhaustion; Hemodynamic instability

N12. a, b, c, e — Low-salt diet and fluid restriction as ordered; Daily weight; report gain > 1–2 kg in a day; Take medications regularly; Report increasing dyspnea or orthopnea